Bézier curve. In turn, we can use these properties to further split the curve by halving each segment along x axis, and computing the exact point on the curve. This point will be one of the corners of each section's bounding box, and we can easily compare the bounding boxes of two curves: they will all be parallel to the cartesian axis, so we just need to check a few inequalities to find out if they intersect.

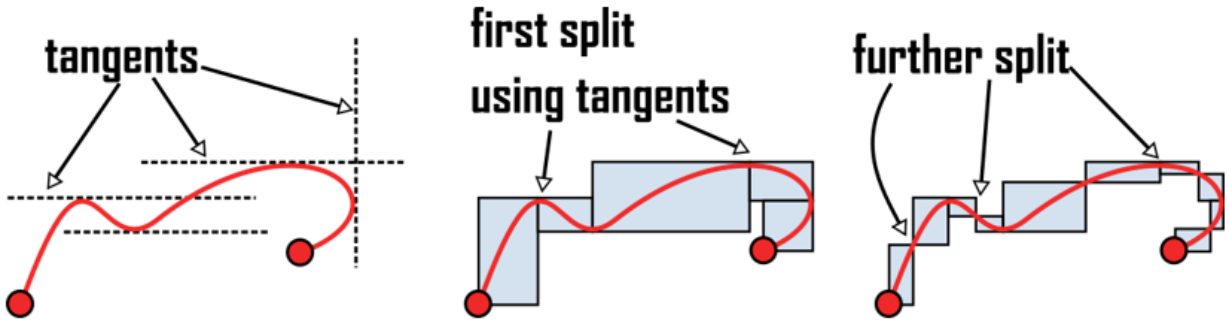


Figure 15.25 The interval subdivision method in action on a generic Bézier curve.

Applying the interval subdivision method to symmetric quadratic Bézier curves is even simpler, because these curves have at most two points where its tangent is parallel to one of the axis; moreover, we can easily compute an initial, coarse-grained bounding box for each of these curves: as shown in figure 15.26, they all lie within the rectangle delimited by the line passing through its endpoints, the parallel to the former tangent to the apex of the curve (at distance $d/2$ from segment SE, as we have previously mentioned), and the perpendicular lines to SE passing to those endpoints.

Therefore, we can informally define the algorithm to check for intersection of two symmetric quadratic Bézier curves as